AP PRECALCULUS

Unit 1 • Lesson 1.7 practice • Answer key

Lesson 1.7 • Rational Functions and End Behavior

4 multiple-choice questions and 1 free-response question • Answer key with explanations

Question 1

What is the horizontal asymptote of $r(x) = \dfrac{5x-1}{x^2+4}$?

(A) $y = 5$

(B) $y = 0$

(C) $y = -\dfrac{1}{4}$

(D) There is none.

Solution

Answer: (B). The degree of the numerator is less than the degree of the denominator.

Question 2

What is the horizontal asymptote of $r(x) = \dfrac{4x^3 - x}{1 - 2x^3}$?

(A) $y = 2$

(B) $y = 4$

(C) $y = -2$

(D) $y = 0$

Solution

Answer: (C). Equal degrees: $\frac{4}{-2} = -2$.

Question 3

What is the slant asymptote of $r(x) = \dfrac{x^2+3x}{x-1}$?

(A) $y = x + 3$

(B) $y = x + 4$

(C) $y = x$

(D) $y = x - 4$

Solution

Answer: (B). $x^2 + 3x = (x-1)(x+4) + 4$, so $r(x) = x + 4 + \dfrac{4}{x-1}$.

Question 4

$\lim\limits_{x \to -\infty} \dfrac{x^3}{x^2+1}$ is

(A) 1

(B) 0

(C) $+\infty$

(D) $-\infty$

Solution

Answer: (D). The quotient behaves like $\frac{x^3}{x^2} = x$, which goes to $-\infty$.

Question 5 (Free response)

Let $r(x) = \dfrac{6x^2+x}{3x^2-12}$ and $s(x) = \dfrac{2x^2-x+3}{x+1}$.

(a) Write limit statements for the end behavior of r.
(b) Find the equation of the slant asymptote of s.
(c) For large positive values of x, is the graph of s above or below its slant asymptote? Explain.

Solution

(a) Equal degrees: $\lim_{x\to\pm\infty} r(x) = \frac{6}{3} = 2$.

(b) Synthetic division by -1 on $2, -1, 3$ gives $2, -3$ and remainder 6: $s(x) = 2x - 3 + \dfrac{6}{x+1}$.
Slant asymptote: $y = 2x - 3$.

(c) For $x > -1$, $\dfrac{6}{x+1} > 0$, so $s(x) > 2x - 3$: the graph is above the asymptote.